

INNOVA JUNIOR COLLEGE
JC 2 PRELIMINARY EXAMINATION 2
in preparation for General Certificate of Education Advanced Level
Higher 2

CANDIDATE
NAME

CLASS

INDEX NUMBER

CHEMISTRY

9647/03

Paper 3 Free Response

22 Sept 2014

2 hours

Candidates answer on separate paper.

Additional Materials: Writing Papers
Data Booklet
Cover Page

READ THESE INSTRUCTIONS FIRST

Write your name and class on all the work you hand in.
Write in dark blue or black pen on both sides of the paper.
You may use a soft pencil for any diagrams, graphs or rough working.
Do not use staples, paper clips, highlighters, glue or correction fluid.

Answer **4 out of 5** questions.
Begin each answer on a fresh sheet of paper.

You are advised to show all working in calculations.
You are reminded of the need for good English and clear presentation in your answers.
You are reminded of the need for good handwriting.
Your final answers should be in 3 significant figures.

At the end of the examination, fasten all your work securely together.
The number of marks is given in the brackets [] at the end of each question or part question.

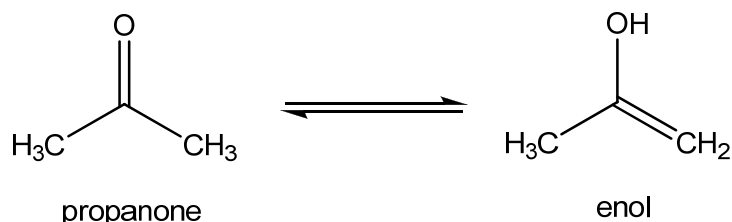
This document consists of **15** printed pages and **1** blank page.



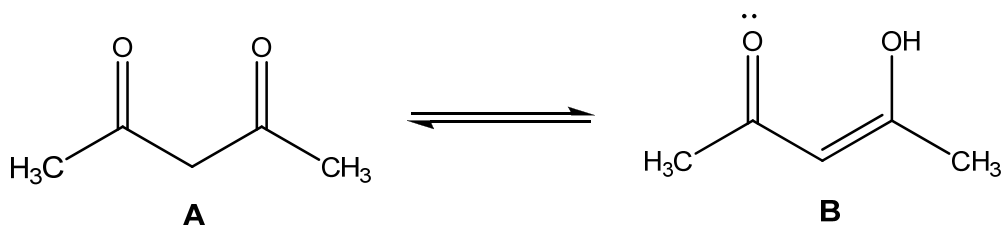
Answer any **four** questions.

- 1 (a) Carbonyl compounds are used as building blocks in the synthesis of many commercially important products, including pharmaceuticals and polymers. Due to the versatile reactivity of the carbonyl compound, they occupy a central role in chemistry.

Carbonyl compounds exist in equilibrium with its isomer, enol through a process called enolization. The reaction involves the transfer of one proton and the shift of the double bond. An example of enolization of propanone is shown below.

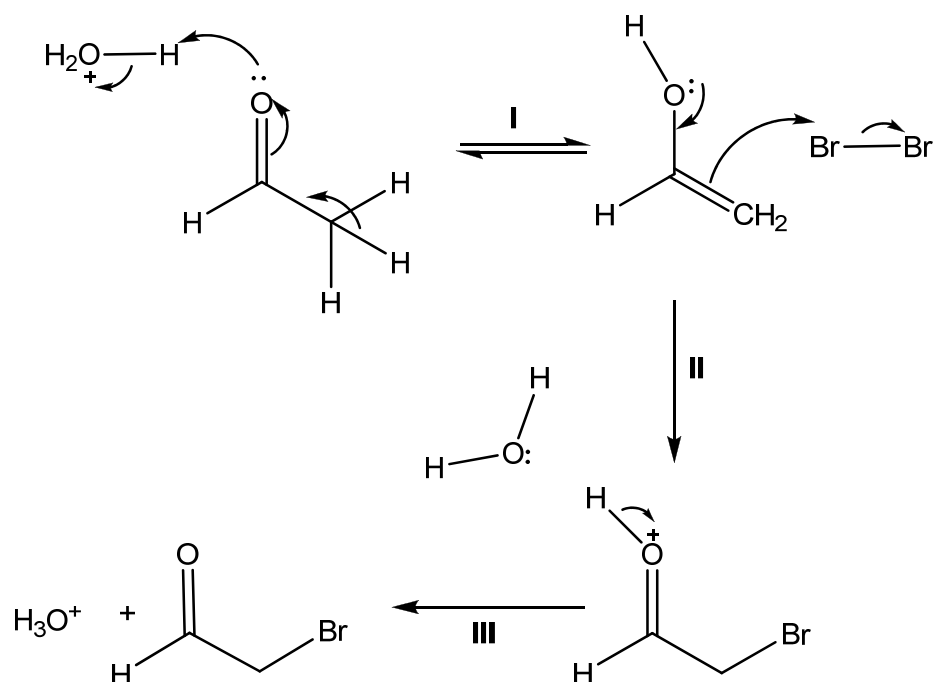


- (i) What is the relationship between the ketone and enol?
- (ii) Write an expression for the equilibrium constant, K_c , for this reaction.
- (iii) The equilibrium constant, K_c , for this reaction is 1.00×10^{-6} . Explain the significance of this value on the equilibrium position and the relative concentrations of the 2 species present at equilibrium.
- (iv) Suggest with the aid of a diagram a reason why the equilibrium constant for the enolization of the diketone **A** shown below is 11.7.



[6]

- (b) Enols are useful compounds in organic chemistry. It allows carbonyl compounds to undergo a variety of reactions. One example is the α -halogenation of carbonyl compounds in the presence of halogen and acid. The reaction is thought to proceed via the following mechanism.



- (i) State the role of the acid in step I.
- (ii) Identify a by-product of the reaction.

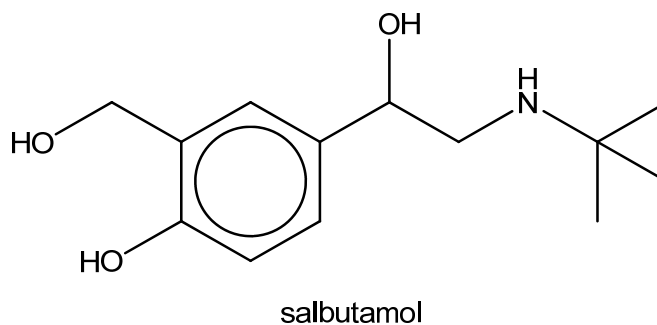
Kinetics experiment shows that the rate equation for the α -halogenation of carbonyl compounds is as follows.

$$\text{Rate} = k[\text{RCHO}][\text{H}^+]$$

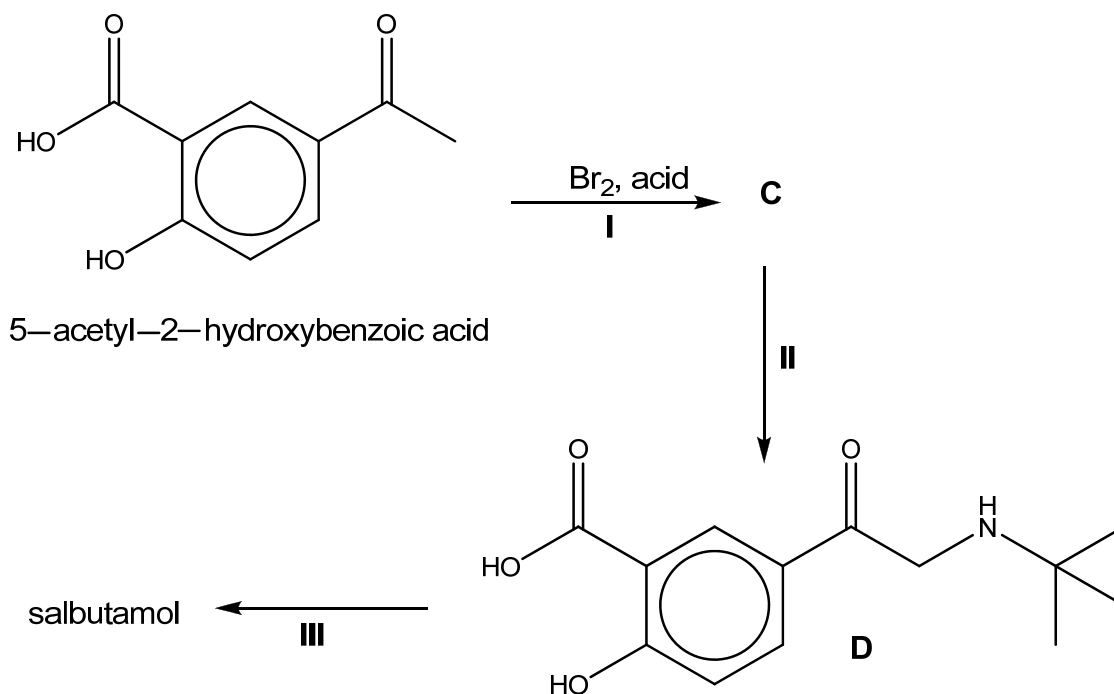
- (iii) How would you expect the rate of the α -halogenation reaction of ethanal with I_2 to compare with that of the above reaction?

[3]

- (c) Salbutamol is a drug used in the treatment of asthma and bronchospasm. The synthesis is relatively straightforward with one of the steps involving the α -halogenation reaction.



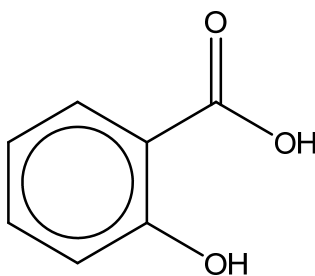
The reaction scheme below outlines the synthesis of salbutamol starting from 5-acetyl-2-hydroxybenzoic acid.



- (i) Suggest the structural formula of compound C.
- (ii) State the reagents and conditions needed for steps II and III.

[3]

- (d) The intermediate 5-acetyl-2-hydroxybenzoic acid used in the synthesis of salbutamol is made from salicylic acid via a 2-step synthesis.



salicylic acid

- (i) Draw the structure of the product formed when salicylic acid reacts with the following:
- Na metal
 - PCl_5
- (ii) The boiling point of the product formed with Na metal in (d)(i) is over 850°C while the boiling point of the product formed with PCl_5 is 206°C . Account for the difference in the boiling points of these compounds.
- (iii) Compare and explain the relative ease of hydrolysis of the product formed with PCl_5 in (d)(i) with chloroethane. State the product formed when aqueous AgNO_3 is added.

[8]

[Total: 20]

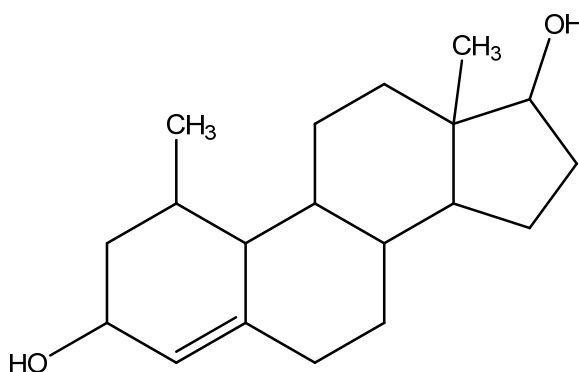
- 2 Manganese is a widely used metal for industrial applications. 10 million tonnes of manganese ores are mined and smelted annually as manganese carbonate. Manganese is also added into alloys of steel to improve its strength. One of the methods of obtaining the metal is by the electrolysis of aqueous manganese(II) sulfate.

(a) During the electrolysis of aqueous manganese(II) sulfate, **two** reactions occur at the inert cathode. A brown solid is also formed in the reaction.

- (i) Construct the two equations for the cathode reactions.
- (ii) Construct an equation for the anode reaction.
- (iii) Calculate the current needed to produce 1 tonne (1×10^6 g) of manganese per day (24 hours).
- (iv) Explain the formation of the brown solid in the reaction and suggest its identity.

[7]

(b) One of the compounds of manganese is potassium manganate(VII) and it is widely used in organic and inorganic chemistry reactions. Hot acidified potassium manganate(VII) can oxidise the following molecule.



- (i) What would be observed during this reaction?
- (ii) State one functional group in the organic product that results from this reaction.
- (iii) Suggest the structure of the organic product.

[3]

- (c) Apart from using metals such as manganese, non-metals such as the halogens and their compounds are also widely used in chemical reactions. One such example is the reaction of the halide ions with concentrated sulfuric acid.

Compare the reactions of solid sodium fluoride, sodium chloride, sodium bromide and sodium iodide with concentrated H_2SO_4 . Include observations and balanced equations with state symbols for any reaction(s) that occur.

[6]

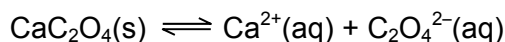
- (d) Chlorine is a reactive halogen. Chlorine gas was mixed with sulfur dioxide in a closed vessel and the system was allowed to reach equilibrium. The product of the reaction is gaseous SO_2Cl_2 .

- (i) Write a balanced equation with state symbols for the reversible reaction and an expression for the equilibrium constant, K_p .
- (ii) Explain the effect on the equilibrium position and K_p if the volume of the container is decreased.

[4]

[Total: 20]

- 3 (a) Oxalic acid, $\text{H}_2\text{C}_2\text{O}_4$, is naturally found in some foods such as spinach, which has 0.97% of oxalic acid by mass. It can combine with metals such as calcium in the body to form calcium oxalate, CaC_2O_4 , a sparingly soluble salt (with K_{sp} value of $1.96 \times 10^{-8} \text{ mol}^2 \text{ dm}^{-6}$ at 25°C). Calcium oxalate, CaC_2O_4 is the main component of kidney stones.



High calcium levels in the urine can also cause kidney stones and a urine calcium test is done to see whether a kidney stone has developed because of high amounts of calcium in the urine. If a person is eating a normal diet, the expected amount of calcium lost in 1 dm^3 of urine is 100 to 300 mg per day.

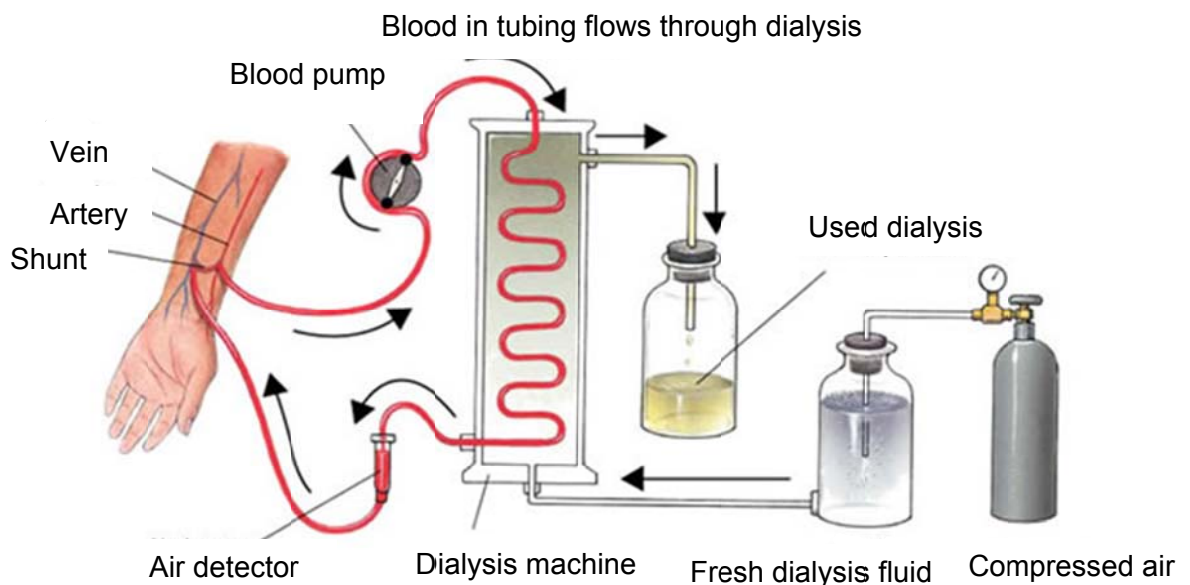
- (i) Write the K_{sp} expression for calcium oxalate.
- (ii) Calculate the concentration of oxalate anion in a saturated solution of calcium oxalate.
- (iii) If a person is eating a normal diet, what would be the minimum required concentration of oxalate ions in order to result in the precipitation of calcium oxalate?
- (iv) Using your answer in (a)(iii), assuming there is no loss of oxalic acid in the body except through the urine, what is the minimum mass of spinach that must be consumed in a normal diet in order to result in the precipitation of calcium oxalate?
- (v) One of the suggested ways to treat mild cases of kidney stones is the consumption of large amount of water. By using Le Chatelier's Principle, explain why this is a feasible treatment.

[8]

- (b) The second law of thermodynamics explains the phenomenon of irreversibility and the increasing entropic trend of nature. The level of entropy during life is described as a U-shaped curve with a trend towards a slower increase in the development phase and a faster increase towards the end of life. Individuals maintain equilibrium until sudden events such as acute illness disrupt the equilibrium. The kidney is an organ dissipating large amounts of energy to maintain the level of entropy of the organism as low as possible.

Research findings have shown that people who have had kidney stones are twice as likely to need dialysis or kidney transplant later in life. Artificial kidney dialysis, such as hemodialysis is used to treat patients suffering from kidney failure. During hemodialysis, a semi-permeable membrane is used to allow waste materials to pass from the blood to the dialysis fluid.

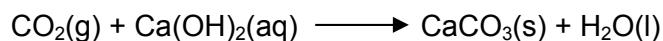
The figure below shows a typical hemodialysis set up.



- (i) Define entropy.
- (ii) Predict and explain the sign of ΔS for the **used** dialysis fluid during the hemodialysis process.

[3]

- (c) A common side effect of kidney disease is the starvation of oxygen in the body. Hence, patient suffering from kidney disease might experience shortness of breath. To measure the rate of respiration, a person needs to exhale into a 2500 cm³ spirometer in which the exhaled air is bubbled at 37 °C and 1 atm through 200 cm³ of 1.00 mol dm⁻³ of calcium hydroxide to obtain a white precipitate of calcium carbonate. The equation for this reaction is



- (i) Given that an average adult exhales about 500 cm³ of air per breath at rest and that exhaled air contains 4% of CO₂ by volume, calculate the amount of CO₂ per breath of air.
- (ii) State an assumption made about CO₂ based on the calculations in (c)(i). Suggest whether this assumption is valid.
- (iii) The white precipitate of calcium carbonate formed in the spirometer was filtered, washed and dried before reacting with 100 cm³ of 0.250 mol dm⁻³ of sulfuric acid. The resultant solution was transferred to a volumetric flask and made up to 250 cm³ with de-ionised water. A 25.0 cm³ aliquot of this diluted solution required 10.0 cm³ of 0.250 mol dm⁻³ sodium hydroxide solution for complete neutralisation.

Calculate the amount of CO₂ present in the exhaled air.

- (iv) Using your answer in (c)(i) and c(iii), calculate the number of times the person exhaled into the spirometer.

[9]

[Total: 20]

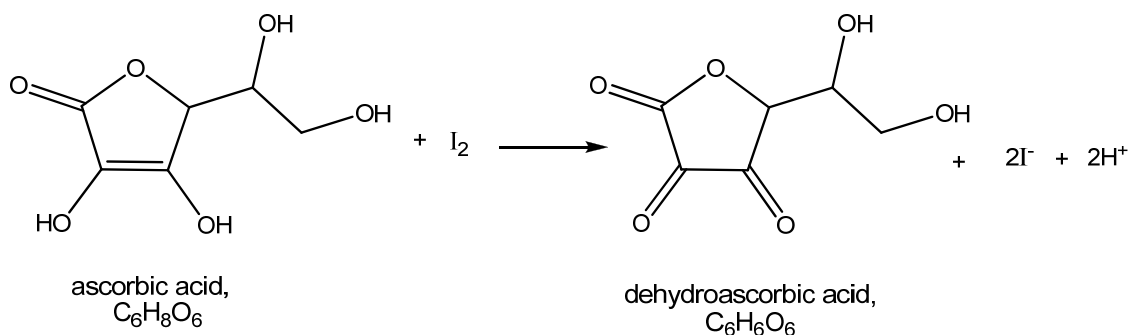
- 4 Iodine was discovered by the French chemist Bernard Courtois in 1811. Courtois was extracting sodium and potassium compounds from seaweed ash. Once these compounds were removed, he added sulfuric acid to further process the ash. He accidentally added too much acid and a violet coloured cloud appeared. The gas condensed on metal objects in the room, creating solid iodine.

Today, iodine is chiefly obtained from deposits of sodium iodate(V), NaIO_3 , and sodium periodate, NaIO_4 , in Chile and Bolivia. Iodine has extensive applications in chemistry and biology.

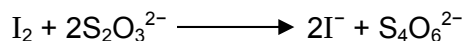
- (a) Draw the structure of IO_4^- .

[1]

- (b) In analytical chemistry, iodometric titration can be used to determine the amount of Vitamin C, also known as ascorbic acid present. Iodine rapidly oxidises ascorbic acid, $\text{C}_6\text{H}_8\text{O}_6$, to produce dehydroascorbic acid, $\text{C}_6\text{H}_6\text{O}_6$.



The amount of excess iodine is determined by titration with standard sodium thiosulfate. The following reaction takes place during the titration.



- (i) Due to the low solubility of iodine in water, it is generated *in situ* using sodium iodate(V), sodium iodide and sulfuric acid.

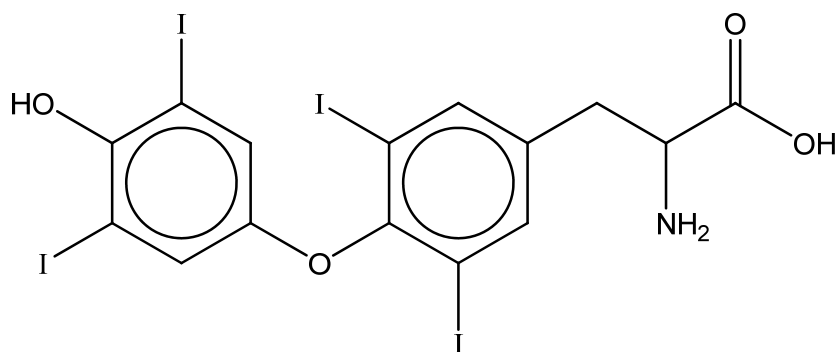
Write a balanced ionic equation for the reaction between IO_3^- and I^- to obtain I_2 .

A 2.5 g sample of Vitamin C tablet is crushed and dissolved in 25 cm^3 of dilute sulfuric acid. To this solution, 0.96 g of sodium iodate(V) and 4.0 g of sodium iodide is added. The resultant solution was then titrated with sodium thiosulfate using starch indicator to determine the amount of excess iodine. 21.60 cm^3 of 0.1 mol dm^{-3} thiosulfate is required for the titration.

- (ii) Calculate the total amount of iodine produced from reacting sodium iodate(V) with sodium iodide.
- (iii) Determine the amount of iodine which reacted with the ascorbic acid.
- (iv) Hence calculate the mass of ascorbic acid in the tablet.

[6]

- (c) In biology, iodine plays a crucial role as constituents of the thyroid hormones, thyroxine and triiodothyronine. Thyroxine, T₄, is formed by the molecular addition of iodine to the amino acid tyrosine while the amino acid is bound to the protein thyroglobulin.



thyroxine, T₄

- (i) Describe what is meant by the tertiary structure of a protein.

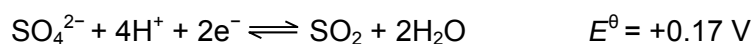
A 19 residue section from a thyroxine (T₄)-containing peptide was digested using two types of enzymes. The enzymatic hydrolysis gave rise to these fragments.

enzyme 1	enzyme 2
val-asn-ala	asn-ile-phe
pro-leu-arg	leu-gln-arg
glu-leu-gln	ala-gln-pro
ile-phe-glu-T ₄	arg-pro-cys-glu
gln-arg	T ₄ -gln-val

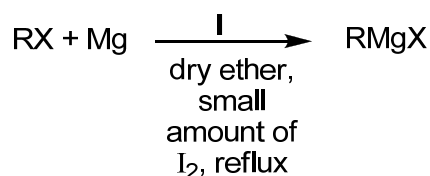
- (ii) Suggest the primary sequence of this 19 residue polypeptide.
- (iii) Besides the use of enzymes, suggest an alternative method to hydrolyse the peptide bonds in a polypeptide.
- (iv) A student wishes to substitute the iodine on the thyroxine with a hydroxyl group. He proposes heating the thyroxine under reflux with aqueous NaOH. Explain why this procedure is unsuitable.
- (v) Draw the structure of the product formed when thyroxine is reacted with aqueous NaOH. (You may assume the R-O-R bond to be inert.)

[7]

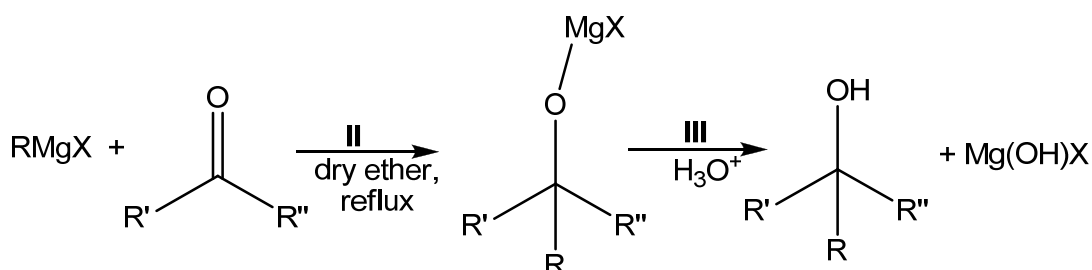
- (d) Iodine can also be used in a galvanic cell. An example is when iodine gas is bubbled into an aqueous solution containing 2.0 mol dm^{-3} potassium iodide solution and this half-cell was connected via a salt bridge, to the following half-cell in standard conditions.



- (i) State the cell notation/ diagram for the above voltaic cell.
- (ii) A student calculated the overall cell e.m.f to be +0.37 V but the overall e.m.f was measured to be +0.29 V. Account for the difference in the values.
- [3]
- (e) In organic chemistry, iodine may be used to catalyse numerous reactions. One example is the Grignard reaction. The Grignard reaction is a very useful reaction in organic chemistry as it allows the formation of carbon-carbon bond. The first stage in the reaction involves the reaction of an alkyl halide with magnesium metal in dry ether to produce the Grignard reagent, RMgX . This reaction is usually slow, hence iodine is added to initiate the process.



The Grignard reagent is then reacted with either an aldehyde or a ketone followed by an acidic quenching to yield the alcohol.

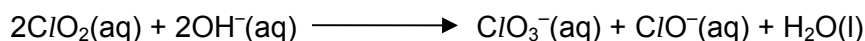


- (i) Suggest why dry conditions are necessary for the formation of the Grignard reagent.
- (ii) Suggest a 3-step synthesis for the formation of $\text{CH}_3\text{CH}_2\text{CH}_2\text{OH}$ using a Grignard reagent.

[3]

[Total: 20]

- 5 (a) Aqueous solution of chlorine dioxide undergoes disproportionation in alkaline solution according to the equation.



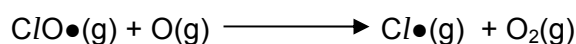
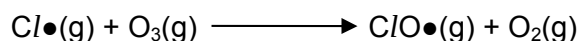
The results of a kinetic study of the disproportionation of aqueous chlorine dioxide in alkaline solution are shown in the table below.

experiment	$[\text{ClO}_2]/\text{mol dm}^{-3}$	$[\text{OH}^-]/\text{mol dm}^{-3}$	initial rate/ $\text{mol dm}^{-3}\text{s}^{-1}$
1	0.0500	0.0216	6.21×10^{-3}
2	0.100	0.0216	2.48×10^{-2}
3	0.0500	0.0648	1.86×10^{-2}

- Determine the orders of reaction with respect to ClO_2 and OH^- .
- Write the rate equation for the reaction.
- Calculate the rate constant, k , of the reaction and state its units.
- Determine the pH of an aqueous solution of chlorine dioxide if its concentration is 0.200 mol dm^{-3} and the initial rate of reaction is $0.145\text{ mol dm}^{-3}\text{s}^{-1}$.

[7]

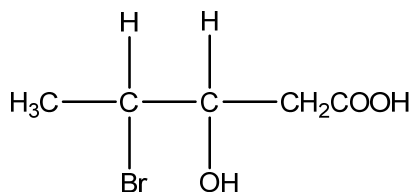
- (b) It is believed that compounds of chlorofluorocarbons are responsible for the depletion of ozone layer via the reactions shown below.



- Explain the role of chlorofluorocarbons in the depletion of the ozone layer.
- Given that the temperature of the stratosphere is -50°C . A student predicted that rate of ozone formation is faster at sea level (15°C). Explain his prediction using the Maxwell Boltzmann distribution curve.
- In fact, practically ozone is not destroyed in the lower atmosphere. Suggest a reason for this.

[5]

- (c) Compound **P**, $C_5H_9ClO_2$ is sparingly soluble in water but readily soluble in aqueous sodium hydroxide. Heating compound **P** in aqueous sodium hydroxide and adding excess acid to the product yields compound **Q**, $C_5H_{10}O_3$, which reacts with warm aqueous alkaline iodine to give a yellow precipitate. Compound **Q** will react to form **R**, $C_5H_8O_2$, when it is refluxed with excess concentrated sulfuric acid. **R** can exhibit cis-trans isomerism. **R** also reacts with aqueous bromine to give a number of products, one of which is shown below:



Deduce the structure of the compounds **P**, **Q** and **R**, explaining the chemistry of the reactions described.

[5]

- (d) With the aid of relevant equations, suggest explanations for the following observations.

Sodium chloride is reacted with an excess of aqueous silver nitrate. The precipitate formed is filtered off and washed with distilled water. The precipitate is then shaken with dilute aqueous ammonia and it dissolves.

If aqueous potassium iodide is used in place of sodium chloride, there is a yellow precipitate.

[3]

[Total: 20]

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